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EXERCISE PHYSIOLOGY

----- Active space --

Significance of exercise :

1. Release of endorphins : mood enhancers.
2. Cardiopulmonary exercise testing (CPET) : To diagnose heart & lung diseases.
3. Lifestyle modifications : Prescribed as treatment for diabetes mellitus.

Type of exercise :

	Isotonic	Isometric
Muscle tension	Constant	Changes
Muscle length	Changes	Constant
Example	Running on a treadmill	Pushing a wall

Sources of energy

00:04:19

1. Phosphocreatinine system/phosphagen system

- ATP from stored creatinine phosphate.
- Duration : 8 – 10 sec (100 m sprints).

2. Glycogen lactic acid system

Anaerobic ATP production

↓

↑ lactic acid

↓

Fatigue

Duration : 1.3 – 1.6 min

3. Aerobic system :

Unlimited energy (ATP) for unlimited time.

Note :

Activities requiring endurance :

Carbohydrate-rich diet → ↑ muscle glycogen storage → Provides energy for up to 4 hrs → Improves endurance

Changes during exercise

00:09:19

SKELETAL MUSCLE CHANGES :

- Contraction of skeletal muscle generates tension.
- Maximum contractile force : 3–4 kg/cm^2.

Blood flow :

3 ml/100g/min (at rest) → 80 ml/100g/min (↑ by 20 times) during exercise.

Organ level blood flow during exercise :

↓ blood flow in renal & splanchnic circulation (GIT)
↑ blood flow to heart and skeletal muscles
→ Redistribution.

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Cerebral circulation : Always maintained constant (D/t autoregulation).

Skin : Initial ↓ → Later ↑ (For heat dissipation).

Exercise hyperemia :

↑ metabolism → Local accumulation of H^+, K^+, lactate → Vasodilation (metabolic end products) → ↑blood flow → Hyperemia

CARDIOVASCULAR CHANGES :

1. Cardiac output (C.O) :

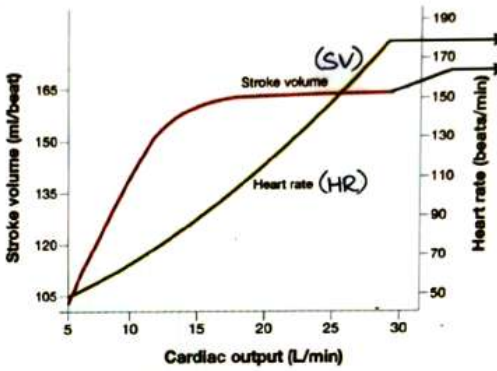
- ↑ 7 fold in athletes.
- ↑ 4 fold in non-athletes.

CO = SV x HR

Constantly increases

Reaches plateau

↑CO d/t ↑HR



2. Anticipatory tachycardia :

Anticipatory ↑ in heart rate even before starting the exercise.

Cause : Joint proprioceptors —Impulse→ Brain.

3. Blood pressure :

	Isotonic	Isometric
Systolic Bp	↑	↑
Diastolic BP	↓	↑
Cause	Local accumulation of metabolic end products → Vasodialation → ↓ TPR & ↓ DBP	Sustained muscle contraction → ↑ Tension → Compresses blood vessels (Arterioles) → ↑TPR & ↑ DBP

TPR : Total peripheral resistance.

DBP : Diastolic blood pressure.

4. Preload :

Sympathetic nervous system stimulation → Releases Norepinephrine → Venoconstriction → ↑ mean systemic filling pressure (MSFP) → ↑ preload

Feedback

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CHANGES IN RESPIRATORY SYSTEM :

----- Active space -----

1. ↑ respiratory rate (RR) & pulmonary ventilation :

Mechanism :

- Exercising skeletal muscles → Release K^+ → Stimulate chemoreceptors → ↑ ventilation.
- ↑ body temp during exercise → ↑ ventilation.

Note :

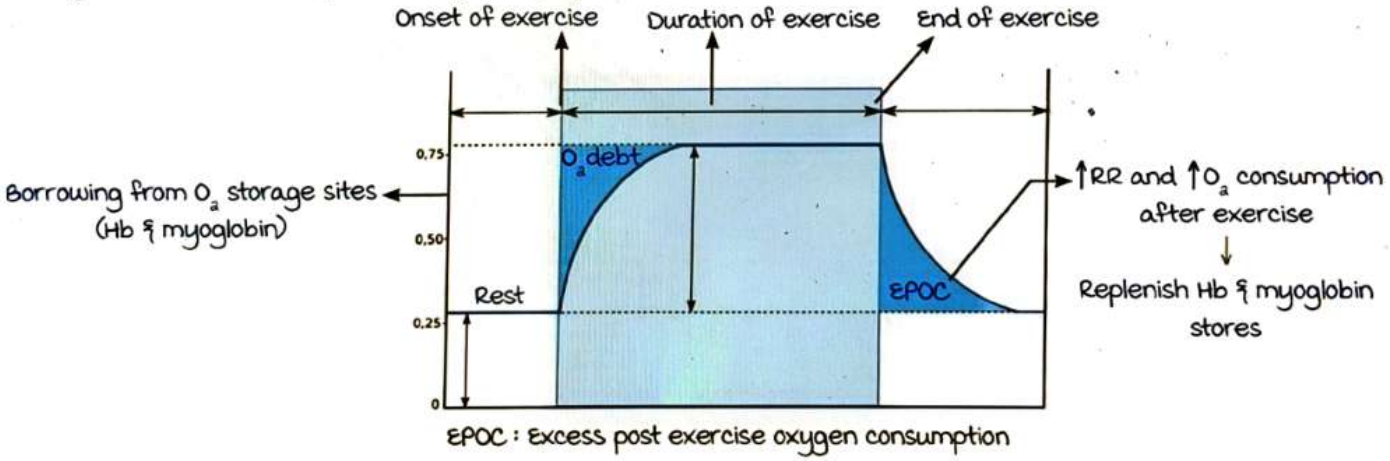
During exercise : Blood gases (pO_a, pCO_a, pH) remain normal.

2. Anticipatory tachypnea :

- ↑ RR before exercise.
- Due to proprioceptors in joints.

3. ↑ Oxygen consumption :

- At rest : 250 ml/min → 1 MET (metabolic equivalent).
- During exercise : ↑O_a consumption → ↑ MET.
- Peak VO_a : O_a consumed at the end of exercise.
- VO_a max : Maximum possible O_a consumption (Theoretical value).



EPOC : Excess post exercise oxygen consumption

4. Diffusion of gases :

- Release of O_a to tissues to meet the demand.
- Oxygen dissociation curve : Shifts to right.

5. Arteriovenous ($A-V\ O_a$) Difference :

- Arteriovenous oxygen difference ↑ during exercise.
- Cause : Tissues extract more O_a.

Effects of smoking

00:36:34

↓ exercise capacity.

↓ in work capacity.

Mechanism :

Toxic chemicals in smoke → • Constriction of airways → • ↑ airflow resistance
• Ciliary paralysis.
• Swelling of epithelial cells.
• ↑ fluid secretion into bronchial tree.



Smokers lung

Feedback

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REGULATION OF BODY TEMPERATURE

Introduction 00:00:24

- Poikilotherm : Body temperature depends on environmental temperature (Eg : Frog).
- Homeotherm : Body temperature constant irrespective of environmental temperature (Eg : Humans).

Normal body temperature :

- 36.8 ± 0.4°C
- 98.2 ± 0.7°F

Changes with circadian rhythm :

- Lowest : 6 AM.
- Highest : 4–6 PM.

Temperature sensors : Transient Receptor Potential (TRP) channels.

For heat :
- TRP V_1 (Vanilloid)
- TRP A_1
- TRP M_2
- TRP M_3

For cold :
- TRP M_8 (menthol)

Unit of heat measurement : Calories.

1 Calorie : Amount of heat required to raise the temperature of 1 Kg of water by 1°C.

Pathology :

- Excessive heat → Protein denaturation → Loss of enzyme function.
- Excessive cold → mechanical tissue damage by ice crystals.

Sources of Heat Generation 00:07:00

metabolism :

Breakdown of food with O_2 consumption → Generates heat (measured by indirect calorimetry).

Respiratory Quotient (RQ) : $\frac{\text{Volume of } CO_2 \text{ produced}}{\text{Volume of } O_2 \text{ consumed}}$

Nutrients	RQ
Carbohydrates	1
Proteins	0.8
Fats	0.7
mixed diet	0.82

Respiratory exchange ratio (RER) : Equal to RQ at steady state.

All the CO_2 produced is expelled by the lungs.

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Mechanism Of Heat Loss

00:10:08 ----- Active space ---

Evaporation : Eg. sweating.

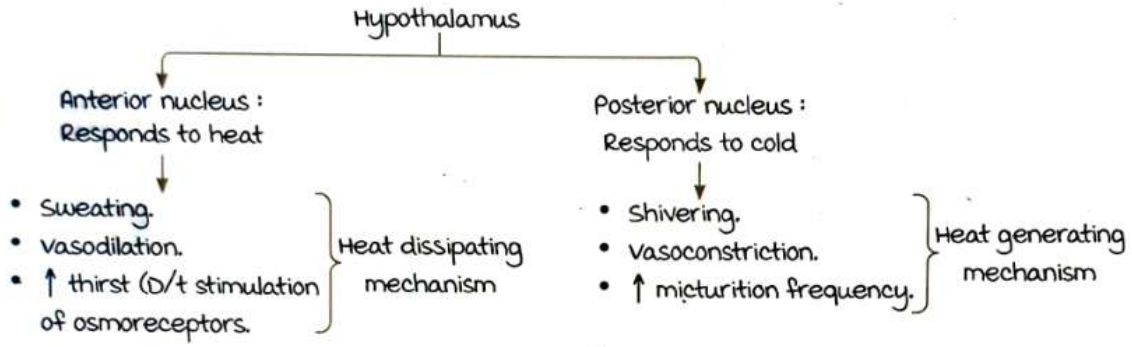
Conduction : Direct transfer of heat to colder objects.

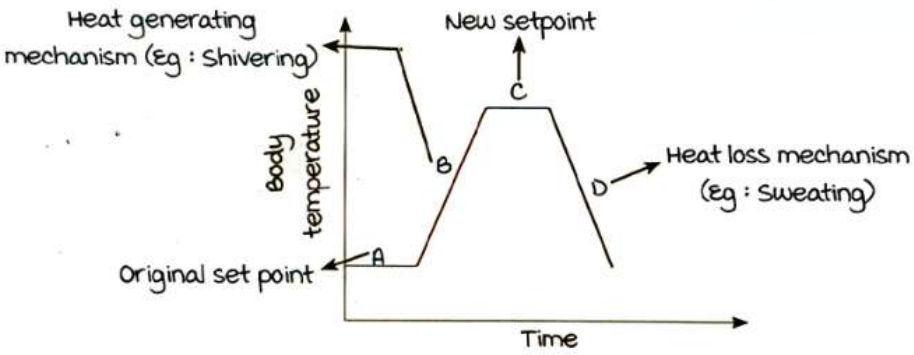
Convection : Loss of heat to air currents.

Radiation : Loss of heat through electromagnetic waves.

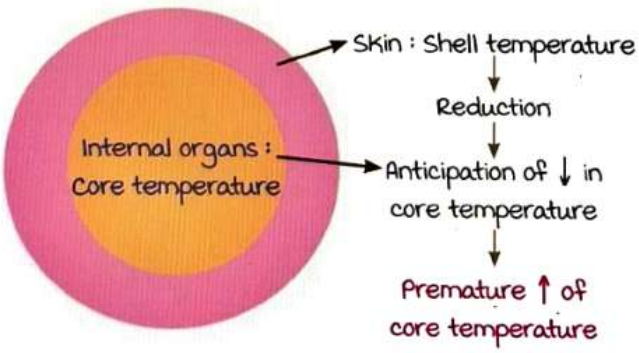
Regulation Of Body Temperature

00:13:35





Feedforward/Anticipatory control :



Fever

00:21:48

Body's defence mechanism enhanced by ↑ in temperature.

Pyrogens :

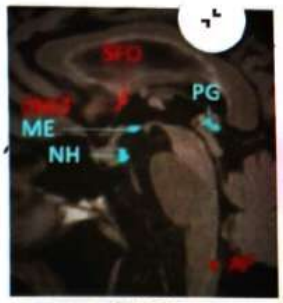
Alteration of hypothalamic set point by pyrogens (Fever-causing agents).

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Pyrogens

- **Exogenous pyrogens**
  - Endotoxins of gram -ve bacteria.
- **Endogenous pyrogens**
  - IL-1.
  - TNF.
  - IL-6.
  - INF-α (used in Rx of hepatitis).



OVLT

Pyrogens cannot cross blood brain barrier (BBB).

- Act on areas outside BBB like organum vasculosum laminae terminalis (OVLT) close to hypothalamic temperature regulation area.
- Produce prostaglandin E_2 $\xrightarrow[EP_3 \text{ receptor}]{}$ ↑ cAMP → Fever, arthritis, myalgia.
- Rx : Antipyretics (↓ PGE2).

Heat Disorders

00:27:10

	Heat cramps	Heat syncope	Heat exhaustion/ Heat collapse	Heat stroke	Malignant hyperthermia
Severity	Very mild	Mild	-	Medical emergency	Most severe
Seen in	Hot environment	Prolongea standing in hot environment	• Classic form : Old age. • Exertion form : Young athletes, soldiers.	• Classic form : Old age. • Exertion form : Young athletes, soldiers.	• Succinyl choline. • Halothane.
Pathology seen	Profuse sweating ↓ Loss of electrolytes	Vasodilation ↓ Venous pooling ↓ ↓ CO, ↓ brain blood flow	Extensive vasodilation ↓ ↓ diastolic filling of heart	Sweating is absent	Overactivity of ryanodine receptor ↓ ↑↑ release of calcium
Presentation	• Muscle pain • Muscle cramps.	• Syncope. • Loss of consciousness.	Loss of consciousness	Triad : 1. Hot dry skin. 2. Coma. 3. Rectal temp >106°F.	Sustained muscle contraction → ↑ heat → ↑ body temperature
Thermoregulatory mechanism	Intact			Not intact	-
Treatment	Electrolyte replacement fluid	Spontaneous recovery on lying down	Lying down + fluid replacement	Active cooling : Cool place, cool water, cold clothes, fluid replacement	Dantrolene sodium (Ryanodine receptor blocker)

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Cold Disorders

00:38:35 ----- Active space -----

HYPOTHERMIA :

Accidental :

- Causes :
  - Cold environment.
  - Drugs : Barbiturates.

Therapeutic :

- Principle : For every 1°C ↓ in body temperature cerebral blood flow ↓ by 7%.
- used to minimize blood loss in neurosurgical procedures.
- used in cardiomyocyte cooling.

Cardiomyocyte cooling :

- Done to prevent death following :

Myocardial infarction → Return of spontaneous circulation (ROSC) → Oxidative stress → Death

Prevention of death by

Cardiomyocyte cooling (Reduce the body temperature by 5°C)

↓ Activates

Reperfusion Injury salvage Kinase (RISK)

↓ Activates

Survival Kinase → Akt phosphorylation ← (+) ← Heat shock protein HSP27 (Positive regulator of Akt)

↓

mitochondrial protection

↓

↑ survival

Feedback